DunesForge - Troubleshooting

Version 0.85 beta. The most common problems, each with what to check first. New to the add-on? Start with Getting Started.

Installation and start-up

There is no DunesForge tab

Press **N** in the 3D Viewport (not in another editor) and look for **DunesForge** on the right edge. If it is missing, open **Edit > Preferences > Add-ons**, find **DunesForge - Dune & Snow Simulator** and make sure it is ticked.

The panel says "RESTART BLENDER"

You installed a new version while Blender was running. Blender keeps the old code in memory: save your work, close Blender and open it again.

Install from Disk shows an error

- The ZIP is about 550 MB. An interrupted download gives an error when installing: download it again. The .sha256 file next to the ZIP lets you check that it is complete.
- Do not unpack the ZIP and do not pack it again yourself; install the file exactly as you received it.
- DunesForge is made for Blender 5.1 or newer. Versions from 4.2 to 5.0 may work, but they have not been thoroughly tested; Blender 4.1 and older cannot install extensions at all.

GPU

"GPU Acceleration" is not ACTIVE

- The GPU path needs an **NVIDIA** card with a current driver. AMD, Intel and Mac computers always run on the CPU, which works but is slower.
- Click **Install CuPy GPU Backend**, wait for the success message, then **restart Blender**.
- The download needs internet access: **Edit > Preferences > System > Network > Allow Online Access**.
- The installer does not install the NVIDIA driver itself. Update it from NVIDIA's website if the message mentions the driver.
- After a Blender update you may need to install the GPU backend again.

The GPU installation stops with a network error

A firewall, proxy or antivirus program must let Blender reach **pypi.org**, where the GPU libraries come from. The error message names a log file; attach it to your report.

A folder called DunesForge appeared in C:\ProgramData

When your Windows user name contains letters such as ł, ó or ż, the GPU installer creates folder links there with plain names, because NVIDIA's compiler cannot read files under such a path. They point to the GPU libraries in your user folder and take no space.

The installation failed or was interrupted

Click **Install CuPy GPU Backend** again. A new installation only becomes active after it passes its tests, so a failed attempt never breaks a working one. **Esc** cancels a running download.

Other add-ons

The GPU worked, then stopped after another add-on was installed

Some add-ons bring their own copy of NVIDIA's CUDA libraries, and Blender keeps the libraries of all extensions in one shared folder. A different version of the CUDA compiler can then end up in DunesForge's GPU backend. A known case is **BlendLuxCore** (the LuxCore renderer). Signs: the panel says **Running on CPU** although the GPU worked before, and tydunes.log mentions `nvrtc`, `__nv_fp8_e8m0` or kernels that do not compile.

1. Disable the other add-on in **Edit > Preferences > Add-ons** and restart Blender.
2. If you need both, use them in different Blender versions, for example the renderer in 4.5 and DunesForge in 5.2. Each Blender version has its own extension folder.
3. Please report it with tydunes.log and the name of the other add-on, so we can list the add-ons that conflict.

The GPU stopped after another add-on or a Blender update changed NumPy

DunesForge's GPU libraries are matched to the NumPy version inside Blender. When another add-on installs a different NumPy, or a Blender update brings one, the panel switches to **Running on CPU**. Click **Install CuPy GPU Backend** again and restart Blender.

An error naming another add-on appears when DunesForge is enabled

Two add-ons tried to use the same internal names; old copies of DunesForge3D (before 2.57) do this. The message names the other add-on. Update it or disable it, then restart Blender.

The simulation does not move

Check these in order:

1. **Did you click Initialize Simulation?** Play does nothing before it.
2. **A red "DUNES FROZEN" box under Play? Freeze Dunes** or **Freeze Terrain** is on, so only the wind and the dust move. Turn both off. The Airborne demo turns them on on purpose.
3. **Is there loose sand?** With **Even sand layer (start)** at 0 and the terrain protected as rock, there is nothing for the wind to move. Add a sand layer, or tick **Terrain itself is loose sand**.
4. **Is Wind Strength above 0?**
5. **Is it just slow?** Dunes move a little per frame. Use **Run Steps** with 300 to 1000 steps to jump ahead.

The result looks wrong

Thick sand breaks into flat steps or terraces along the wind

Since 0.85.4 the default settings no longer do this. It comes from **Crest Shear Ceiling** and **Saltation Feedback (Owen)** switched on together on a thick layer (about 1 m or more). Set both back to 0, or set **CROSS-SLOPE TRANSPORT** to **0.1**; a red box next to the sand layer points at it. Older scenes saved before 0.85.4 may still have both on.

Piles grow too steep, into spikes or towers

Where sand keeps arriving (a **Sand Source**, sand blown against a rock), slopes can stand far steeper than real sand. In **Physics** (Interface: **Advanced**) set **Settle Slopes Every N** to **1** and **Settle Rounds** to **10** (up to **20** if needed): the slopes then stay at the natural angle of about 34 degrees for roughly a quarter to a third more time per step. **Avalanche Trigger** fixes it the natural way, with real slides, but costs more. Details: Notes for beta testers.

Jagged pixel steps where bare rock meets sand

Under **Surface Shader**, set **Rock Edge Smoothing** to about **1.0**. It is meant for larger areas of bare rock; leave it at 0 when the rock appears only as tiny specks.

The dunes are tiny, or there is barely any relief

The simulation needs time: run more steps. Also check **Even sand layer (start)**: a thin layer only makes small dunes.

Sand slides off the mountains

Steep terrain cannot hold a thick loose layer: sand slides down to its natural angle. Lower the terrain's **Image Amplitude** or **Height Scale**, or protect the slopes as rock.

Water

Shallow water -- the flowing-water simulation and the demos marked **(Experimental)** -- is a preview. It needs deeper integration with the rest of the simulator and will probably not be part of the first release (1.0); it is planned for a version after it. The simple water level, rain and moisture are not affected.

I cannot see the water

- Flowing water needs a water surface: click **Create Water Plane** in the **Water / Moisture** panel.
- Very shallow water (puddles) can look like wet sand. Lower **Opacity Depth** (for example to 0.05).
- If the **Water Level** sits below the terrain, there is nothing to fill.

A saved water demo has no water after reopening

The flowing water is part of the running simulation and is not stored in the .blend file; the file keeps only the water the demo started with. Click **Initialize Simulation** and that water comes back.

The water simulation is slow

Waterfalls and steep channels are the most expensive. Use a lower **Water Detail**, and remember that one Play frame can be one second of water time.

Speed

- Lower the **Resolution**: 256 is fine for trying ideas, 512 to 1024 for final results.
- Use **Run Steps** instead of Play: it calculates many steps and updates the viewport once.
- Use **Add Fast Surface (cheap)** instead of the full shader while working, and **Solid** view while simulating.
- Some demos take a few minutes to build (avalanches, the pond, the Grand Scenes). The status bar shows progress; the view updates when they are done.

Materials and rendering

The terrain is grey, flat or black

- Switch the viewport to **Material Preview** or **Rendered**.
- Click **Add Surface Shader** under **3. Surface appearance**.
- **Flat Colour (diagnostic)** checks whether the geometry itself is right.
- **Black only in Cycles, on Blender older than 5.2?** That Blender cannot compile the rain shader together with procedural detail or texture bombing. Pick **2 - Rain + scans** and click **Build This Variant**, or use Blender 5.2 or newer. A red box under the variant list warns when the material will not compile.

Interface

A control from the manual is missing

The **Interface** switch at the top of the panel is probably on **Simple**. Switch to **Standard** or **Advanced**. Hidden controls keep their values and still affect the result.

Ctrl+Z does not undo the sand

That is expected: Blender's undo cannot see the simulation. Use the DunesForge **Undo** button, or **Ctrl+Alt+Z** (redo: **Ctrl+Shift+Alt+Z**).

A scene behaves strangely after opening

Click **Check Terrain** to see what is wrong, then **Repair Scene State**. **Reset Simulation** returns the terrain to its starting shape.

Reporting a problem

Please send:

1. The version from the top of the panel (for example **0.85.5 beta**) and your Blender version.
2. Your graphics card, and whether the panel says **GPU Acceleration: ACTIVE**.
3. What you did step by step, what you expected and what happened.
4. A screenshot, plus the log file. Click **Open Log Folder** (main panel) and attach **tydunes.log**. The folder is Blender's configuration folder, for example `C:\Users\NAME\AppData\Roaming\Blender Foundation\Blender\5.1\config\tydunes`.

If you can, attach a small scene that shows the problem.